

Project Design Phase-II Technology Stack (Architecture & Stack)

Date	7 March 2026
Team ID	PNT2022TMIDxxxxxx
Project Name	Malaria Detection using Deep Learning
Maximum Marks	4 Marks

Technical Architecture:

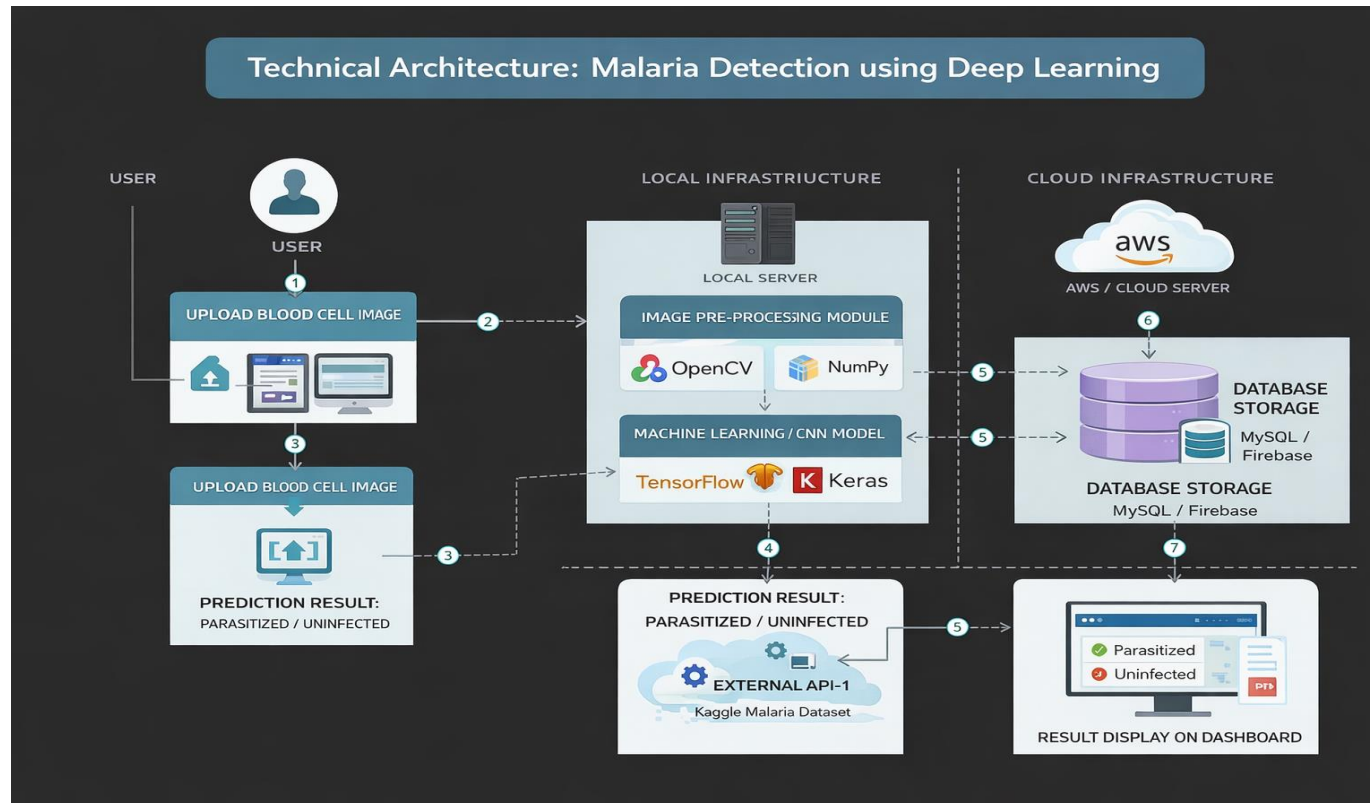


Table- : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface where users upload blood smear images	HTML, CSS, JavaScript
2	Application Logic-1	Handles image upload and processing	Python
3	Application Logic-2	Image preprocessing and feature extraction	OpenCV, NumPy
4	Application Logic-3	Malaria parasite detection using trained model	TensorFlow / Keras
5	Database	Stores detection results and user data	MySQL / SQLite
6	Cloud Database	Optional storage for remote access	Firebase / AWS
7	File Storage	Stores uploaded blood smear images	Local Storage / Cloud Storage
8	External API-1	Image processing support	OpenCV Library
9	External API-2	Dataset access for training	Kaggle Malaria Dataset
10	Machine Learning Model	Detects malaria infected cells	CNN Model
11	Infrastructure (Server / Cloud)	Deployment environment	Local Server / AWS Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for development	TensorFlow, Keras, OpenCV
2	Security Implementations	Protects system and user data	Password Encryption, HTTPS
3	Scalable Architecture	System can handle multiple users	3-Tier Architecture
4	Availability	System accessible anytime through web	Cloud Hosting
5	Performance	Fast image processing and prediction	GPU Processing, Optimized Model

